

How to find real solution of system of differential equations (SDE) with complex eigenvalue/eigenvectors?

Answer:

When you encounter complex eigenvalues, it signifies that the system has a rotational component (spirals or centers). The challenge is to convert the complex-valued solution you find into the two real-valued solutions needed for a general real solution.

The Core Idea

For a real matrix $\mathbf{A}$, complex eigenvalues always come in conjugate pairs: $\lambda = \alpha \pm i\beta$. You only need to work with one of these pairs (e.g., $\alpha + i\beta$) and its corresponding complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$.

1. You will form one complex solution: $\mathbf{x}(t) = \mathbf{v} e^{\lambda t}$.
2. Using Euler's formula ($e^{i\theta} = \cos(\theta) + i \sin(\theta)$), you will expand this complex solution.*
3. The expansion will naturally separate into a real part and an imaginary part:
 $\mathbf{x}(t) = \mathbf{x}_{\text{re}}(t) + i \mathbf{x}_{\text{im}}(t)$ where
 $\mathbf{x}_{\text{re}}(t) = e^{\alpha t} [\mathbf{p} \cos(\beta t) - \mathbf{q} \sin(\beta t)]$ and $\mathbf{x}_{\text{im}}(t) = e^{\alpha t} [\mathbf{p} \sin(\beta t) + \mathbf{q} \cos(\beta t)]$
4. $\mathbf{x}_{\text{re}}(t)$ and $\mathbf{x}_{\text{im}}(t)$ are a fundamental set of **real solutions** to the original system.
5. The general real solution is then $\mathbf{x}(t) = c_1 \mathbf{x}_{\text{re}}(t) + c_2 \mathbf{x}_{\text{im}}(t)$

*In practice, students may skip step 2, and use the formulas in step 3 to write down $\mathbf{x}_{\text{re}}(t)$ and $\mathbf{x}_{\text{im}}(t)$ in terms of α , β , $\mathbf{p}$ and $\mathbf{q}$.

Interpreting the Solution for a Phase Portrait

The form of the eigenvalue $\lambda = \alpha \pm i\beta$ tells you the shape of the phase portrait:

- **α (the real part):** Determines stability.
 - $\alpha < 0$: **Stable Spiral (Sink)**. Trajectories spiral into the origin (Asymptotically stable)
 - $\alpha > 0$: **Unstable Spiral (Source)**. Trajectories spiral away from the origin (Unstable)
 - $\alpha = 0$: **Center**. Trajectories are closed ellipses or circles. In our example, $\alpha = 0$, so the phase portrait consists of concentric ellipses. (Neutrally stable)
- **β (the imaginary part):** Determines the frequency of rotation. A larger $|\beta|$ means faster rotation.

Let's solve the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ for the matrix: $\mathbf{A} = \begin{pmatrix} 1 & -5 \\ 1 & -1 \end{pmatrix}$

Step 1: Find the Eigenvalues

The eigenvalues are a complex conjugate pair $\lambda = 0 \pm 2i$. Here, $\alpha = 0$ and $\beta = \pm 2$.

Step 2: Find One Eigenvector

We only need to find the eigenvector for one of the eigenvalues. Let's use $\lambda = 2i$ so that $\alpha = 0$ and $\beta = 2$. Then the corresponding eigenvector $\mathbf{v}$ is: $\mathbf{v} = \begin{pmatrix} 1 + 2i \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Here $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

Step 3: Find $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$

With the $\alpha = 0$ and $\beta = 2$ from step 1 and $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ from step 2, we have

- $\mathbf{x}_{re}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos(2t) - \begin{pmatrix} 2 \\ 0 \end{pmatrix} \sin(2t) = \begin{pmatrix} \cos(2t) - 2 \sin(2t) \\ \cos(2t) \end{pmatrix}$
- $\mathbf{x}_{im}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \sin(2t) + \begin{pmatrix} 2 \\ 0 \end{pmatrix} \cos(2t) = \begin{pmatrix} \sin(2t) + 2 \cos(2t) \\ \sin(2t) \end{pmatrix}$

Step 4: Write the General Real Solution

These two vector functions, $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$, are our two fundamental set of real solutions. The general solution is a linear combination of them:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} \cos(2t) - 2 \sin(2t) \\ \cos(2t) \end{pmatrix} + c_2 \begin{pmatrix} \sin(2t) + 2 \cos(2t) \\ \sin(2t) \end{pmatrix}$$

** Alternative eigenvalue

It is also possible to use the other (conjugate) eigenvalue $\lambda = -2i$ so that $\alpha = 0$ and $\beta = -2$. This will give slightly different expressions for $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$, and hence the general solution.

Step 2': Find the other Eigenvector

If we use $\lambda = -2i$, then the corresponding eigenvector $\mathbf{v}$ will be the conjugate of the eigenvector above: $\mathbf{v} = \begin{pmatrix} 1 - 2i \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} -2 \\ 0 \end{pmatrix}$. So $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$.

Step 3': Find $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$

With the $\alpha = 0$ and $\beta = -2$ and $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$ from step 2', we have

- $\mathbf{x}_{re}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos(-2t) - \begin{pmatrix} -2 \\ 0 \end{pmatrix} \sin(-2t) = \begin{pmatrix} \cos(2t) - 2 \sin(2t) \\ \cos(2t) \end{pmatrix}$
- $\mathbf{x}_{im}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \sin(-2t) + \begin{pmatrix} -2 \\ 0 \end{pmatrix} \cos(-2t) = \begin{pmatrix} -\sin(2t) - 2 \cos(2t) \\ -\sin(2t) \end{pmatrix}$

Note that the new $\mathbf{x}_{re}(t)$ is the same as the one using $\lambda = 2i$; while the new $\mathbf{x}_{im}(t)$ is the negative (a scalar multiple) of the one using $\lambda = 2i$.

Step 4': Write the General Real Solution

$$\mathbf{x}(t) = c_1 \begin{pmatrix} \cos(2t) - 2 \sin(2t) \\ \cos(2t) \end{pmatrix} + c_2 \begin{pmatrix} -\sin(2t) - 2 \cos(2t) \\ -\sin(2t) \end{pmatrix}$$

Note that the negative signs in the second term can be "absorbed" into the parameter c_2 in the general solution.

Scenario 2

Let's solve the system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ for the matrix: $\mathbf{A} = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}$

Step 1: Find the Eigenvalues

The eigenvalues are a complex conjugate pair $\lambda = -1 \pm 2i$. Here, $\alpha = -1$ and $\beta = \pm 2$.

Step 2: Find One Eigenvector

We only need to find the eigenvector for one of the eigenvalues. Let's use $\lambda = -1 + 2i$ so that $\alpha = -1$ and $\beta = 2$. Then the corresponding eigenvector $\mathbf{v}$ is: $\mathbf{v} = \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Here $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Step 3: Find $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$

With the $\alpha = -1$ and $\beta = 2$ and $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ from step 2, we have

- $\mathbf{x}_{re}(t) = e^{-t} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos(2t) - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin(2t) \right] = e^{-t} \begin{pmatrix} \cos(2t) \\ -\sin(2t) \end{pmatrix}$
- $\mathbf{x}_{im}(t) = e^{-t} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin(2t) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos(2t) \right] = e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix}$

Step 4: Write the General Real Solution

These two vector functions, $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$, are our two fundamental set of real solutions. The general solution is a linear combination of them:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} \cos(2t) \\ -\sin(2t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix}$$

** Alternative eigenvector of the same eigenvalue

There are different ways to solve for the eigenvectors. For example, using MATLAB, you are likely to get the eigenvector $\mathbf{v} = \begin{pmatrix} -i \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ for the same eigenvalue. In this case, $\mathbf{p}$

$= \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$. Note that the new eigenvector differs from the original one above by a scalar of $-i$: $\begin{pmatrix} -i \\ 1 \end{pmatrix} = -i \begin{pmatrix} 1 \\ i \end{pmatrix}$. So it is still a valid eigenvector for the same eigenvalue $\lambda = -1 + 2i$.

Then

- $\mathbf{x}_{\text{re}}(t) = e^{-t} \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos(2t) - \begin{pmatrix} -1 \\ 0 \end{pmatrix} \sin(2t) \right] = e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix}$
- $\mathbf{x}_{\text{im}}(t) = e^{-t} \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin(2t) + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \cos(2t) \right] = e^{-t} \begin{pmatrix} -\cos(2t) \\ \sin(2t) \end{pmatrix}$

Note that the new $\mathbf{x}_{\text{re}}(t)$ is the same as the original $\mathbf{x}_{\text{im}}(t)$; while the new $\mathbf{x}_{\text{im}}(t)$ is the negative of the original $\mathbf{x}_{\text{re}}(t)$.

The general solution $\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -\cos(2t) \\ \sin(2t) \end{pmatrix}$ is slightly different but equivalent to the original one.